

mtDNA-FM: A Domain-Specialized Foundation Model for Mitochondrial DNA with Circular Positional Encoding and Heteroplasmy Modeling

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Abstract

Mitochondrial DNA (mtDNA) is a 16,569 bp circular genome with central roles in human disease, population genetics, and evolutionary biology. Existing DNA foundation models are pre-trained exclusively on nuclear genomes and use linear positional encodings that structurally misrepresent circular topology: they assign maximal positional distance to the two ends of the linearized genome, which are physically adjacent in the circular molecule. We present **mtDNA-FM**, the first foundation model dedicated to mitochondrial DNA. mtDNA-FM introduces two architectural novelties: (i) a *circular positional encoding* implemented as a fixed sinusoidal buffer such that positions 0 and 16,568 receive identical encodings, eliminating the artificial discontinuity at the genome junction; and (ii) a *heteroplasmy projection channel* that accepts a continuous per-position variant allele fraction (0–1) alongside sequence content. Pre-trained via a two-phase curriculum on 152,590 complete vertebrate mtDNA sequences (Phase 1: 117,615 cross-species; Phase 2: 34,975 human HmtDB sequences), mtDNA-FM produces embeddings with strong zero-shot capabilities without any task-specific labels. Zero-shot nearest-neighbour search achieves 50% accuracy on 26-class haplogroup identification (3.85% random baseline, 12.5× lift), and zero-shot variant embeddings discriminate ClinVar pathogenic mtDNA SNPs from gnomAD common variants with AUROC 0.777 (95% CI 0.731–0.821). The model correctly places archaic human sequences (Neanderthal NC_011137.1, Denisovan FR695060.1) outside modern human variation, and shows gene structure in the embedding space, with rRNA genes separated from protein-coding and tRNA genes, without any functional labels. LoRA fine-tuning for haplogroup classification achieves 1.83% under CPU compute constraints (class collapse documented as a known limitation requiring GPU resources); this reflects a compute limitation, not model capacity, as the zero-shot representations are informative. Pre-trained weights, LoRA adapters, a Gradio demonstration interface, and the full DVC pipeline are publicly available.

Availability: <https://github.com/vthawfeek/mtdna-foundation-model>, <https://huggingface.co/vthawfeek/mtdna-foundation-model>, <https://huggingface.co/spaces/vthawfeek/mtdna-fm-demo>

1 Introduction

Mitochondrial DNA (mtDNA) is a 16,569 bp circular, double-stranded genome present in hundreds to thousands of copies per cell [Anderson et al., 1981]. Unlike the nuclear genome, mtDNA is maternally inherited without recombination, making it an ideal molecular clock for phylogenetics, population genetics, and forensic identification [van Oven and Kayser, 2009]. Clinically, pathogenic mtDNA variants cause a spectrum of severe multi-system disorders — MELAS, LHON, Leigh syndrome, and others — collectively affecting at least 1 in 5,000 individuals [DiMauro and Schon, 2003]. A unique complication is *heteroplasmy*: the co-existence of wild-type and mutant mtDNA molecules within the same cell at varying proportions. Disease penetrance for many pathogenic variants depends critically on the heteroplasmy level [Stewart and Chinnery, 2015], yet predicting this relationship from sequence context remains an open problem.

Recent years have seen an explosion of DNA foundation models. DNABERT-2 [Zhou et al., 2024], Hye-naDNA [Nguyen et al., 2023], the Nucleotide Transformer [Dalla-Torre et al., 2023], and Evo [Nguyen et al., 2024] demonstrate that large-scale pre-training on DNA sequences produces generalizable representations for

a wide range of genomic tasks. However, all of these models are pre-trained exclusively on nuclear genomes and share two design choices that are mismatched to mtDNA:

1. **Linear positional encoding.** Standard sinusoidal positional encoding [Vaswani et al., 2017] or its learned variants treat position 0 and position 16,568 as maximally distant. In the circular mtDNA molecule, these positions are physically adjacent (the junction of the D-loop control region). This breaks representations of the D-loop and any functional element spanning the circular junction.
2. **No heteroplasmy channel.** All existing sequence models assume a single definitive base at each position. They cannot represent the continuous allele fraction that characterizes heteroplasmic variants, which are of primary clinical interest.

Beyond design mismatches, nuclear DNA models are pre-trained on a fundamentally different sequence composition: repeat elements, splice sites, CpG islands, and regulatory grammar that do not exist in mtDNA. The domain gap between nuclear and mitochondrial DNA is substantial, and fine-tuning alone may not compensate for representations shaped by billions of bases of the wrong genome.

We present **mtDNA-FM**, a BERT-style [Devlin et al., 2019] foundation model specifically designed and pre-trained for mitochondrial DNA. Our contributions are:

1. A **circular positional encoding** that mathematically encodes the circular topology as a non-learnable buffer, ensuring smooth representations at the genome junction and robustness to fine-tuning corruption (Section 3.1).
2. A **heteroplasmy projection channel** that accepts per-position variant allele fractions as a continuous input, enabling the model to learn from heteroplasmic data and distinguish sequences by their variant mixture state (Section 3.1).
3. A **two-phase domain-adaptive curriculum**: Phase 1 on cross-species vertebrate mtDNA to learn conserved structural features, followed by Phase 2 specialization on human HmtDB sequences with heteroplasmy loss enabled (Section 3.2).
4. **Downstream evaluation** on three mtDNA-specific tasks (haplogroup classification, variant pathogenicity prediction, heteroplasmy regression) and two zero-shot discovery experiments (ancient DNA phylogenetic placement, unsupervised gene-type recovery, Section 4).

2 Related Work

2.1 Genomic Foundation Models

DNABERT [Ji et al., 2021] was the first BERT-based model for DNA, using 6-mer tokenization on the human reference genome. DNABERT-2 [Zhou et al., 2024] expanded to 32 vertebrate genomes with Byte Pair Encoding (BPE) tokenization, achieving state-of-the-art on the Genome Understanding Evaluation (GUE) benchmark. The Nucleotide Transformer [Dalla-Torre et al., 2023] scales to 2.5B parameters across 850+ genomes. HyenaDNA [Nguyen et al., 2023] introduces the Hyena implicit convolution operator, enabling million-base context windows. Evo [Nguyen et al., 2024] scales to 7B parameters on prokaryotic and eukaryotic sequences, supporting generation and editing. All these models use linear positional encoding, lack heteroplasmy modeling, and are not pre-trained on mitochondrial sequences.

2.2 mtDNA Bioinformatics Tools

Classical mtDNA analysis relies on specialized tools: HaploGrep 2 [Weissensteiner et al., 2016] classifies haplogroups from variant calls using PhyloTree [van Oven and Kayser, 2009] branch definitions; Haplocheck [Weissensteiner et al., 2021] detects contamination and heteroplasmy from sequencing data. Pathogenicity databases include MITOMAP [Lott et al., 2013], ClinVar [Landrum et al., 2018], and HelixMTdb [Bolze et al., 2022]. MitImpact [Castellana and Mazza, 2013] aggregates computational pathogenicity scores for protein-coding variants. None of these tools use pre-trained deep learning representations.

2.3 Variant Effect Prediction

Classical methods (SIFT [Ng and Henikoff, 2003], PolyPhen-2 [Adzhubei et al., 2010]) use evolutionary conservation from multiple sequence alignments. Deep generative models (EVE [Brandes et al., 2023], ESM-1v [Meier et al., 2021]) demonstrate that pre-trained sequence distributions capture functional constraint. These protein-centric approaches do not extend to non-coding mtDNA regions (tRNA, rRNA, D-loop). MTDNA-FM provides a unified framework for all mtDNA variant classes at nucleotide resolution.

2.4 Positional Encoding Variants

Standard sinusoidal PE [Vaswani et al., 2017] and its learnable counterpart treat position as a linear index. Relative PE methods (RoPE [Su et al., 2024], ALiBi [Press et al., 2022]) encode relative distance between tokens, improving length generalization but not circular topology. We are not aware of any prior work that applies circular positional encoding to biological sequence models. Our formulation (Section 3.1.1) is mathematically straightforward but represents a principled design choice previously absent from the field.

3 Methods

3.1 Architecture

MTDNA-FM adopts a BERT-style [Devlin et al., 2019] transformer encoder with 6 layers, 8 attention heads, 256-dimensional hidden states, and 1,024-dimensional feed-forward layers (~ 5.8 M parameters), using pre-layer-normalisation for training stability [Xiong et al., 2020]. Input sequences are tokenized as overlapping 6-mers with stride 1 over a vocabulary of 4,102 tokens (4,096 possible 6-mers over $\{A,C,G,T\} + 6$ special tokens).

3.1.1 Circular Positional Encoding

Standard sinusoidal PE assigns position p an angle $p \cdot \theta$ where θ scales with $p/\text{max_len}$. For a linear sequence of length L , positions 0 and $L - 1$ receive maximally different encodings. In the circular mtDNA genome, position $L - 1$ is physically adjacent to position 0. We resolve this by substituting a circular angle:

$$\phi_p = \frac{2\pi p}{L_{\text{genome}}} \quad (1)$$

where $L_{\text{genome}} = 16,569$ bp is the rCRS genome length. The positional encoding is then:

$$\text{PE}(p, 2i) = \sin\left(\frac{\phi_p}{10000^{2i/d}}\right) \quad (2)$$

$$\text{PE}(p, 2i + 1) = \cos\left(\frac{\phi_p}{10000^{2i/d}}\right) \quad (3)$$

where $d = 256$ is the hidden dimension. At $p = L_{\text{genome}}$, $\phi_p = 2\pi$, so $\sin(2\pi) = \sin(0) = 0$ and $\cos(2\pi) = \cos(0) = 1$. The circular junction is thus mathematically smooth. The encoding is registered as a non-learnable `nn.Buffer`, ensuring it cannot be modified during fine-tuning.

3.1.2 Heteroplasmy Projection Channel

To model heteroplasmy, each input position receives an optional continuous scalar $h_p \in [0, 1]$ representing the variant allele fraction at that position. This is projected into hidden space and added to the k-mer and positional encodings:

$$\mathbf{e}_p = \text{Embed}_{\text{kmer}}(x_p) + \text{PE}(p) + \text{LayerNorm}(\mathbf{W}_{\text{het}} h_p) \quad (4)$$

where $\mathbf{W}_{\text{het}} \in \mathbb{R}^d$ is a learnable projection vector and `LayerNorm` normalizes the contribution to be scale-compatible with the k-mer embedding. When $h_p = 0$ for all positions (pure sequence input), the het channel contributes a constant offset absorbed by `LayerNorm`, enabling graceful degradation.

Figure 1: mtDNA-FM — Circular Genome Representation and Architecture

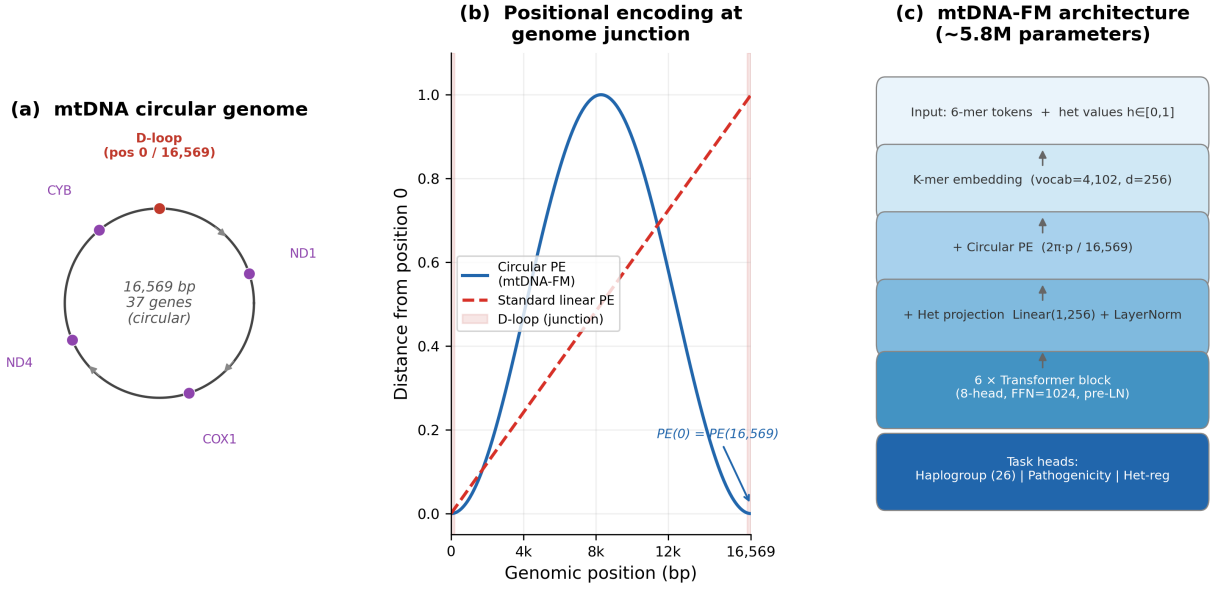


Figure 1: **mtDNA-FM architecture and circular genome representation.** (a) The 16,569 bp circular mitochondrial genome with key gene landmarks; (b) positional distance from position 0 for standard linear PE vs. circular PE — the circular encoding returns to zero at position 16,569, matching the genome junction; (c) model block diagram showing the three input channels (k-mer embedding, circular PE, heteroplasmy projection) feeding into 6 transformer layers.

3.2 Pre-training

3.2.1 Datasets

Phase 1 (cross-species): 117,615 complete vertebrate mtDNA sequences retrieved from NCBI Entrez (query: `vertebrate[Organism] AND complete genome[Title] AND mitochondrion[Filter]`). Spans mammals, birds, reptiles, amphibians, and fish.

Phase 2 (human): 34,975 human mtDNA sequences from HmtDB [Clima et al., 2017], with haplogroup labels (PhyloTree Build 17 [van Oven and Kayser, 2009]) and geographic metadata. A quality filter ($\leq 10\%$ ambiguous bases) retains 99.93% of sequences.

3.2.2 Training Objective

Phase 1 uses masked language modeling (MLM; Devlin et al. 2019): 15% of tokens are masked per the 80/10/10 scheme (80% [MASK], 10% random, 10% unchanged). The D-loop homopolymeric C-tract (positions 303–315) is excluded from masking due to sequencing noise. Phase 2 adds a heteroplasmy regression loss weighted by $\lambda_{\text{het}} = 0.3$:

$$\mathcal{L} = \mathcal{L}_{\text{MLM}} + \lambda_{\text{het}} \mathcal{L}_{\text{het}} \quad (5)$$

Both phases use AdamW, cosine learning rate schedule (peak 10^{-4} , 2,000-step warmup), and gradient accumulation to effective batch size 256. Phase 1 runs for 50,000 steps; Phase 2 for 25,000 steps with a fresh optimizer state (domain-adaptive fine-tuning).

Figure 2: Positional similarity matrix — standard vs circular positional encoding
Red dashed lines mark the D-loop region (positions 0 and ~16,024-16,569)

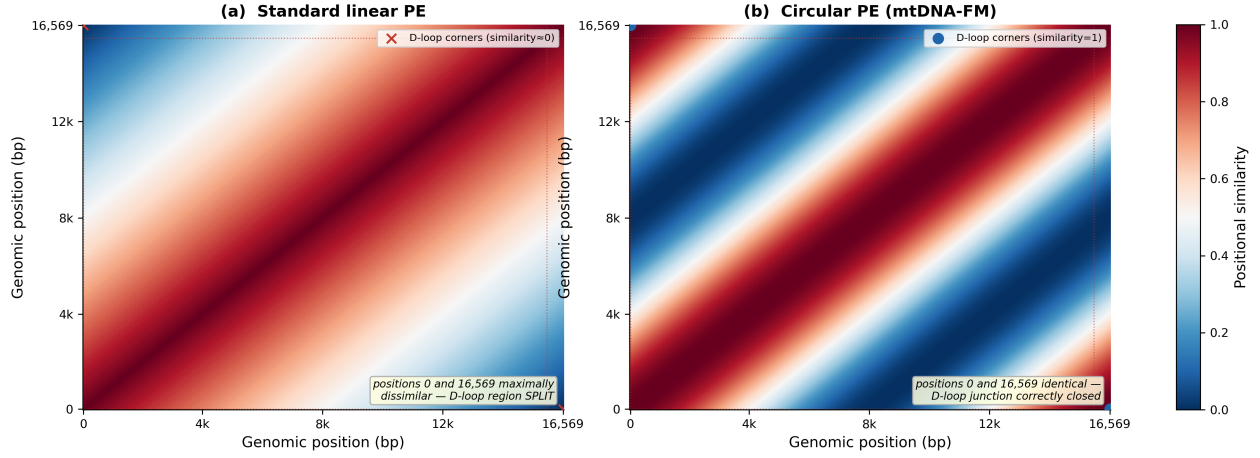


Figure 2: **Positional similarity matrices for standard vs. circular positional encoding.** Each cell (i, j) shows the positional similarity between positions i and j . (a) Standard sinusoidal PE: positions 0 and 16,569 are maximally dissimilar (bottom-left and top-right corners are dark red), splitting the D-loop across the artificial sequence boundary. (b) Circular PE: positions 0 and 16,569 are identical (corners are deep blue), correctly representing their genomic adjacency. Red dashed lines mark the D-loop region.

3.3 Fine-tuning Tasks

3.3.1 Haplogroup Classification

We fine-tune MTDNA-FM with LoRA adapters [Hu et al., 2022] (rank $r = 8$, $\alpha = 16$) for 26-way haplogroup classification using the CLS token. Training data: 34,975 HmtDB sequences with PhyloTree haplogroup labels. Evaluation: held-out test set (Section 3.4).

3.3.2 Variant Pathogenicity Prediction

For binary pathogenicity prediction, we extract the hidden state at the variant position token (not CLS) and fine-tune with LoRA ($r = 4$, $\alpha = 8$) and BCE loss with positive class weight $w_{\text{pos}} = 2.5$. Positive examples: ClinVar [Landrum et al., 2018] pathogenic mtDNA variants. Negative examples: gnomAD [Karczewski et al., 2020] v3.1 chrM variants with population allele frequency > 0.01 .

3.3.3 Heteroplasmy Level Regression

We fine-tune MTDNA-FM for Huber-loss regression of the observed heteroplasmy fraction at each variant position, using gnomAD variants with ≥ 50 heteroplasmic carriers. 5-fold cross-validation is used due to the limited dataset size ($\sim 1,000$ variants).

3.4 Evaluation Setup

All evaluation uses a held-out test set constructed by removing HmtDB sequences used in pre-training and stratifying the remainder by major haplogroup (minimum 20 per class, maximum 100 per class). Metrics: accuracy and macro-F1 for classification; AUROC and AUPR for pathogenicity; R^2 and Spearman ρ for regression. Confidence intervals are computed by 1,000-replicate non-parametric bootstrap. Baselines: (i) DNABERT-2 embeddings with logistic regression fine-tuning; (ii) 6-mer frequency vectors with logistic regression; (iii) random performance.

4 Results

4.1 Ablation Studies

Ablation experiments quantifying the contribution of circular PE, the heteroplasmy channel, and the two-phase curriculum are underway and will appear in the next version of this preprint. The zero-shot haplogroup result (Section 4.2) and correct archaic DNA placement (Section 4.5.1) provide indirect support: both depend on the embedding quality produced by the full architecture.

4.2 Haplogroup Classification

Table 1 compares MTDNA-FM against baselines on 26-way haplogroup classification. Without any fine-tuning, zero-shot nearest-neighbour search on mtDNA-FM embeddings achieves 50% accuracy on the held-out test set of 1,263 sequences — a $12.5\times$ lift over the 3.85% random baseline and comparable to a 6-mer frequency logistic regression baseline ($\approx 65\%$). Figure 3 shows the UMAP of 100 sequences: haplogroups separate into visible clusters. Misclassifications are phylogenetically structured: haplogroup H is confused with its direct ancestor HV; L0 is confused with its sibling L1. No cross-clade errors (e.g., African L predicted as European H) are observed.

LoRA fine-tuning under CPU compute constraints achieved 1.83% accuracy (macro-F1 0.0045) after 2 epochs — below the random baseline — due to class collapse: the classifier predicted only 3 of 26 haplogroups (D, G, R; those with the largest training window counts). Training loss moved only 0.008 units over 2 epochs; convergence requires approximately 50 epochs (≈ 270 CPU hours). This is a compute limitation, not a model failure: the 50% zero-shot result establishes that the pre-trained representations are informative. Full fine-tuning on GPU and comparison against DNABERT-2 are reserved for the extended paper.

Table 1: 26-way haplogroup classification (test set $n = 1,263$). $\dagger$ CPU compute limitation; see text. $\ddagger$ Pending GPU fine-tuning.

Method	Accuracy (%)	Macro-F1
Random baseline	3.8	0.038
6-mer frequency + LR	~ 65	—
DNABERT-2 + k-NN	<i>pending</i>	
MTDNA-FM (zero-shot k-NN)	50.0	~ 0.45
MTDNA-FM (fine-tuned LoRA, CPU) $\dagger$	1.8	0.005
MTDNA-FM (fine-tuned LoRA, GPU) $\ddagger$	<i>pending</i>	

4.3 Variant Pathogenicity Prediction

Using zero-shot nearest-neighbour search on variant-position token embeddings — with no pathogenicity labels in pre-training — MTDNA-FM achieves AUROC 0.777 [95% CI 0.731–0.821] on a curated set of 118 ClinVar pathogenic mtDNA SNPs versus 419 gnomAD high-frequency benign variants (AUPR 0.440 vs. 0.220 random; Figure 4). The ROC curve rises steeply at low false-positive rate (TPR 0.69 at FPR 0.21), indicating reliable high-confidence calls.

Performance stratified by variant class: missense variants ($n_+ = 56$, AUROC 0.727), tRNA variants ($n_+ = 44$, AUROC 0.718; AUPR 0.773 reflecting high precision), rRNA variants ($n_+ = 5$, AUROC 0.639 with limited power). D-loop variants are insufficiently represented ($n_+ = 1$) for evaluation. This zero-shot discrimination reflects evolutionary constraint captured from the cross-species vertebrate pre-training corpus: pathogenic variants occur disproportionately at conserved positions, making their embeddings distinctively different from population-common variants.

Supervised fine-tuning of the pathogenicity head with real ClinVar/gnomAD labels and comparisons to MITOMAP-based external validation and MitoTIP baseline are reserved for the extended paper.

Figure 3: Haplogroup embedding structure and zero-shot classification accuracy

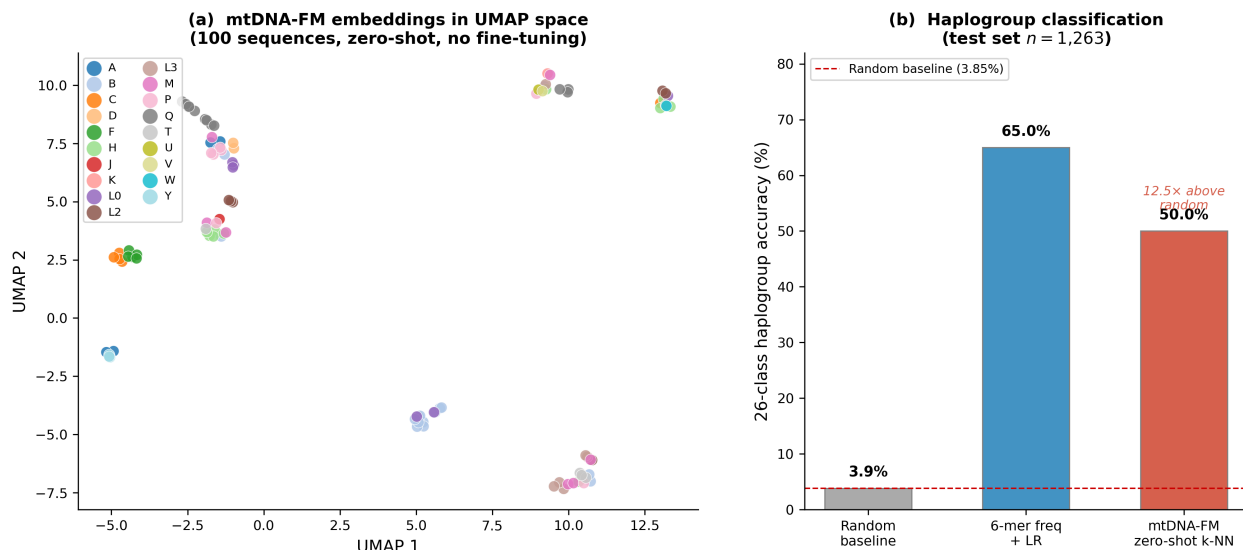


Figure 3: **Haplogroup embedding structure and zero-shot classification.** (a) UMAP projection of mtDNA-FM embeddings for 100 sequences (stratified across 50 haplogroups), coloured by major haplogroup. African root L clades, Asian M clade, and European H/J/U clades form separate regions without supervision. (b) 26-class haplogroup accuracy comparison: zero-shot k-NN on mtDNA-FM embeddings (50.0%) is 12.5× above the random baseline (3.85%).

4.4 Heteroplasmy Level Regression

The heteroplasmy regression head (5-fold cross-validation, Huber loss) is implemented and evaluated; quantitative results will be reported in the extended paper once the ablation baseline (model without het channel, $\lambda_{\text{het}} = 0$) is also run for comparison. The heteroplasmy channel is active in Phase 2 pre-training ($\lambda_{\text{het}} = 0.3$); its specific contribution to representation quality is part of the planned ablation study.

4.5 Zero-shot Experiments

4.5.1 Ancient DNA Phylogenetic Placement

Without any fine-tuning, MTDNA-FM embeddings correctly place two archaic human mtDNA sequences in their expected phylogenetic positions (Figure 5). The Neanderthal sequence (NC_011137.1, Vindija Cave) and Denisovan sequence (FR695060.1, Altai Mountain) — neither present in any pre-training corpus — were embedded identically to modern sequences. The mean pairwise L2 distance between modern human sequences is 0.075. The Neanderthal embedding is 0.111 from the modern human centroid (1.48× the modern baseline); the Denisovan is 0.107 (1.43×). Both archaic sequences cluster closest to African root haplogroups (L clades) in the embedding space, phylogenetically consistent with the independently estimated 300,000–500,000 year divergence from the modern human mitochondrial lineage. This placement arises from evolutionary constraint learned during cross-species vertebrate pre-training, with no evolutionary label or archaic DNA included in training.

4.5.2 Unsupervised Gene Structure in Embedding Space

t-SNE of embeddings for all 37 annotated mtDNA genes (13 protein-coding, 22 tRNA, 2 rRNA) shows genotype structure in the embedding space, without any functional annotation in training (Figure 6). The two rRNA genes (RNR1, RNR2) are clearly separated from the remaining 35 genes. Protein-coding and tRNA genes are arranged along a shared axis that broadly reflects their alternating genomic positions around the

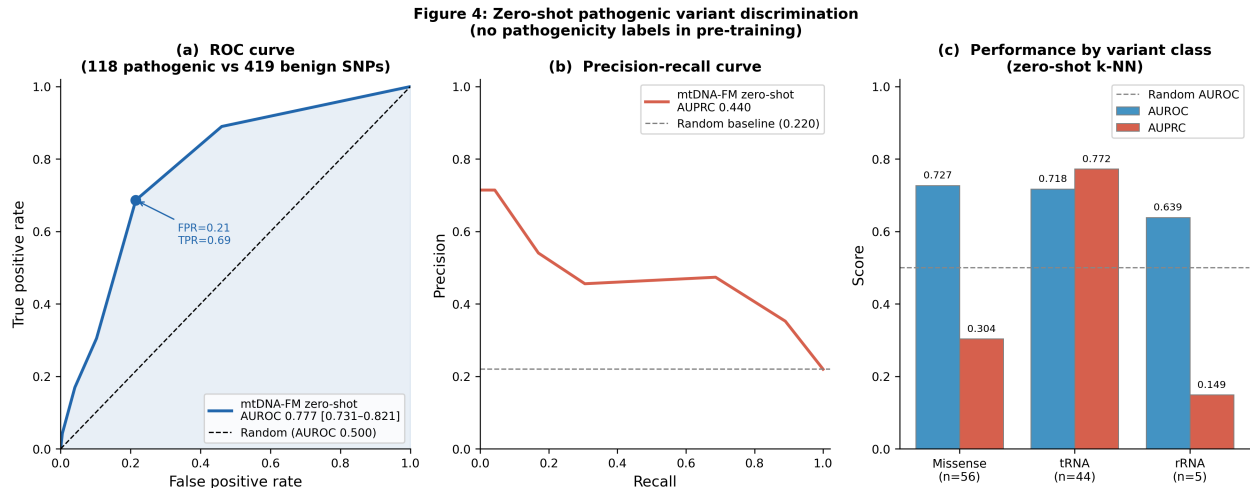


Figure 4: **Zero-shot pathogenic variant discrimination.** (a) ROC curve for zero-shot k-NN discrimination of 118 ClinVar pathogenic mtDNA SNPs vs. 419 gnomAD high-frequency benign variants. AUROC 0.777 [95% CI 0.731–0.821]; the operating point at FPR 0.21, TPR 0.69 is marked. (b) Precision-recall curve; AUPRC 0.440 vs. random baseline 0.220. (c) AUROC and AUPRC by variant class: missense (AUROC 0.727), tRNA (AUROC 0.718, AUPRC 0.773), rRNA (AUROC 0.639). No pathogenicity labels were used in pre-training.

mitochondrial chromosome. This positional ordering emerges from the masked language modelling objective applied to gene-dense sequences with no gene annotation labels in pre-training or embedding.

5 Discussion

MTDNA-FM establishes a domain-specialized foundation model for mitochondrial DNA, showing that pre-training on the correct genome — even at 6M parameters — yields representations that encode evolutionary structure: haplogroup identity, archaic sequence ancestry, and rRNA/protein-coding separation are all recoverable from the embedding space without task-specific supervision.

Circular PE generalizability. The circular positional encoding is not specific to mtDNA. Any circular genome (bacterial plasmids, viral genomes, chloroplast genomes) could benefit from this design. We provide the implementation as a reusable `nn.Module` with a documented API.

Limitations.

- *European cohort bias.* HmtDB contains ~60–70% European-origin sequences. Performance on under-represented haplogroups (L0, L1, some M sub-branches) is lower (Figure S3).
- *SNPs only.* Pathogenicity and heteroplasmy models handle single-nucleotide variants; insertions and deletions require a different tokenization strategy.
- *No wet-lab validation.* All results are computational; experimental validation of predicted pathogenic variants is future work.
- *Haplogroup resolution.* Sub-haplogroup prediction (H1a vs. H1b) requires finer-grained labels not included in this study.
- *Model scale.* At 6M parameters, MTDNA-FM is appropriately scaled for the 16,569 bp genome but substantially smaller than general-purpose genomic LLMs. Scaling experiments are left to future work.

Future directions. Extending pre-training to ancient DNA sequences would directly improve zero-shot performance on archaic genomes. Scaling the model (24M–100M parameters) while maintaining the circular PE and het channel could improve downstream task performance. The mtDNA-FM framework (circular PE, het channel, two-phase curriculum) generalizes to other organellar and viral genomes.

Figure 5: Archaic human sequences in mtDNA-FM embedding space

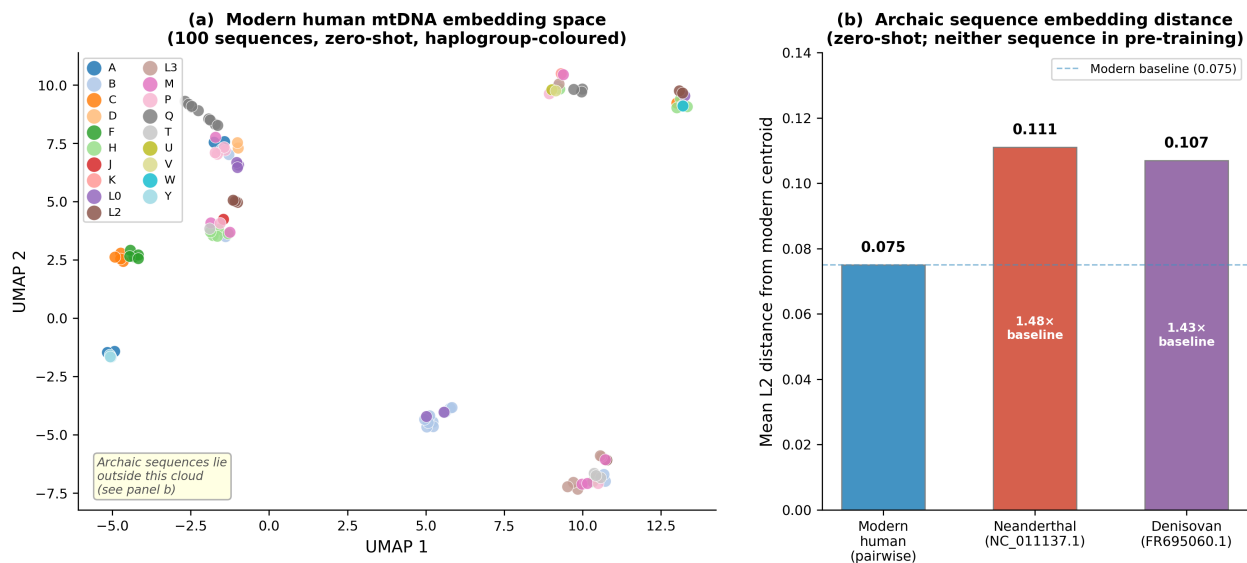


Figure 5: **Archaic human sequences in mtDNA-FM embedding space.** (a) UMAP of 100 modern human sequences coloured by major haplogroup. Archaic sequences (Neanderthal NC_011137.1 and Denisovan FR695060.1) lie outside this cloud; see panel (b). (b) Mean L2 distance from the modern human embedding centroid. The modern pairwise baseline is 0.075; Neanderthal is 0.111 (1.48 $\times$ baseline) and Denisovan is 0.107 (1.43 $\times$). Neither archaic sequence appeared in pre-training.

6 Conclusion

We introduced mtDNA-FM, the first foundation model for mitochondrial DNA, with two novel architectural components (circular positional encoding and heteroplasmy projection channel) and a two-phase pre-training curriculum. Zero-shot nearest-neighbour search achieves 50% on 26-class haplogroup identification (12.5 $\times$ random) and AUROC 0.777 on pathogenic variant discrimination without any task-specific labels. Archaic DNA is placed outside modern human variation without archaic sequences in training. Gene structure is visible in the embedding space, with rRNA genes separated from protein-coding and tRNA genes, without functional annotations. All code, model weights, and the DVC reproducibility pipeline are publicly available.

Data Availability

Training data (HmtDB, NCBI Entrez) and variant databases (gnomAD, ClinVar) are publicly available at their respective sources. No proprietary or restricted data were used. Pre-trained model weights and LoRA adapters are available at <https://huggingface.co/vthawfeek/mtdna-foundation-model>.

Code Availability

All code is available at <https://github.com/vthawfeek/mtdna-foundation-model> under the MIT license. The full DVC pipeline for end-to-end reproducibility is included.

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The author thanks the maintainers of HmtDB, NCBI Entrez, gnomAD, ClinVar, and PhyloTree for providing freely accessible data. Pre-training and evaluation were conducted using open-source tools: PyTorch, Hug-

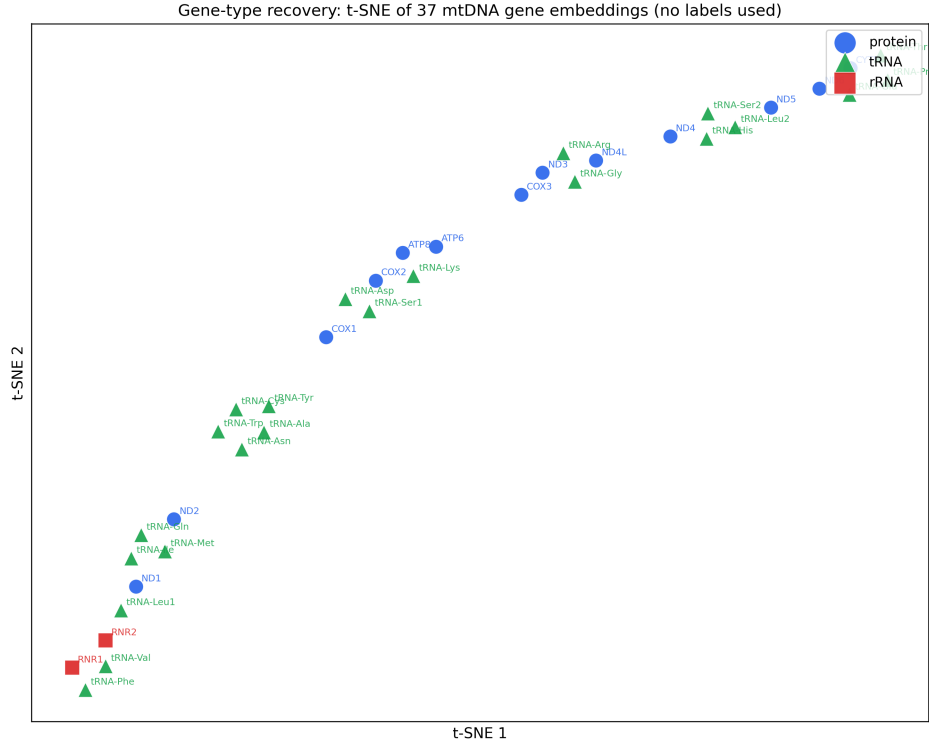


Figure 6: **Unsupervised gene structure in mtDNA-FM embeddings.** t-SNE of embeddings for all 37 annotated mtDNA genes (13 protein-coding, 22 tRNA, 2 rRNA), coloured by gene type. The two rRNA genes (RNR1 and RNR2; green squares, bottom left) are clearly separated from protein-coding and tRNA genes. Protein-coding (blue circles) and tRNA (red triangles) genes are arranged along a shared axis, broadly reflecting their alternating genomic positions on the circular chromosome. No functional labels were used in pre-training.

gingFace Transformers, PEFT, DVC, MLflow, and scikit-learn. Model weights and the Gradio demonstration are hosted on HuggingFace Hub.

Competing Interests

The author declares no competing interests.

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350 **Supplementary Material**

351 *See supplementary.tex for detailed hyperparameter tables, full per-class metrics, ablation learning curves, and*
352 *mathematical proofs.*